KINGFAISALUNIVERSITY·CCSIT·EMBEDDEDSYSTEMS

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MACHINE GUARDIAN

NON-INTRUSIVEINDUSTRIALACCESSCONTROL&ENERGYMONITORING·ESP32IOTSYSTEM

ABSTRACT

Machine Guardian is an integrated IoT security system that restricts industrial equipment access to NFC-authenticated personnel while monitoring real-time energy consumption. The ESP32 microcontroller implements an event-driven state machine, a "Silencing Protocol" for EMI-free ADC sampling, and ambient-baseline cancellation for accurate net current measurement. Data is visualised on a locally-hosted web dashboard with live charts, Excel export, and automated Telegram alerts — requiring no cloud infrastructure.

TEAMMEMBERS

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NFC ACCESS CONTROL

The PN532 module reads MIFARE ISO14443A cards. UIDs are validated against an employee table stored in the ESP32's NonVolatile Storage (NVS), persisting through power cycles. Employees can be added, removed, or deactivated live from the web dashboard.

IDLE Scan card
AUTH 30 s window
RUNNING Current detected

NON-INTRUSIVE SENSING

The SCT-013 current clamp attaches directly to the machine's power cable — no wiring modifications, no downtime. The CT outputs 50 mA at 100 A primary, converted by a 33 Ω burden resistor to a voltage readable by the ESP32 ADC

```
// True RMS over 2000 samples
rmsV = √(Σ(samples²) / N)
A = (rmsV / 33Ω) × 2000
```

LIVE WEB DASHBOARD

Served directly from ESP32 PROGMEM — no SD card, no external server. The Chart.js dashboard shows live current draw with ambient and threshold reference lines, runtime by day, operator usage pie chart, and a full event log with timestamps.

```
GET /api/status
GET /api/power
POST /api/set_threshold
```

AMBIENT CANCELLATION

Room wiring induces 4–8 A of apparent current even with no machine running. The system captures a 30-sample baseline with the machine off, adds a 10% safety margin, and subtracts it from every reading — reducing the effective noise floor to <0.3 A net.

```
rawAmps − gAmbient = netAmps
// EMA filter α = 0.15
out = α·net + (1−α)·prev
```

TELEGRAM ALERTS

On unauthorized access, the ESP32 sends an HTTPS POST directly to api.telegram.org using WiFiClientSecure — no intermediary server required. The supervisor receives an instant alert with timestamp and location. All compute stays on-device.

⚠ UNAUTHORIZED ACCESS
Machine: started without valid card
Time: 2026-04-30 · 10:14
Location: Machine Guardian

OFFLINE EXCEL EXPORT

SheetJS builds a multi-sheet .xlsx file entirely in the browser — Power Log, Runtime by Day, Operator Summary, and Event Log. Demo data is excluded from all sheets. The file contains only real machine sessions with timestamps and net amperage

Sheet 1 · Power Log (raw + net A, W)
Sheet 2 · Runtime by Day
Sheet 3 · Operator Summary
Sheet 4 · Event Log

EVENT-DRIVEN STATE MACHINE

IDLE
Red LED on
Awaiting NFC scan
Current monitored

→

AUTHENTICATED
Yellow LED on
Valid card scanned
30 s to start machine

→

RUNNING
Green LED on
Net current ≥ threshold
Power & runtime logged

⤵

UNAUTHORIZED
Red flash + buzzer
Telegram alert sent
Runtime tracked separately